

1.3 Exchanging data	
How data is exchanged between different systems	
1.3.1 Compression, Encryption and Hashing	a) Lossy vs Lossless compression. b) Run Length Encoding and dictionary coding for lossless compression. c) Symmetric and asymmetric encryption. d) Different uses of hashing.
1.3.2 Databases	a) Relational database, flat file, primary key, foreign key, secondary key, normalisation and indexing. b) Methods of capturing, selecting, managing and exchanging data. c) Normalisation to 3NF. d) SQL - Interpret and modify (list of key words). e) Referential Integrity. f) Transaction processing, ACID (Atomicity, Consistency, Isolation, Durability), record locking and redundancy.
1.3.3 Networks	a) The TCP/IP Stack. b) Protocol layering. c) LANs and WANs. d) Packet and circuit switching. e) Protocols. f) Client-server and Peer to peer.
1.3.4 Web Technologies	a) HTML, CSS and JavaScript. b) Search engine indexing. c) PageRank Algorithm. d) Server and client side processing.